Deck 05 · Optimization II: constraints, multipliers, and the KOMO formulation

4.1 Constraints and KKT: the fence, the force, the multiplier

4.2 Inverse kinematics and trajectories as one nonlinear program

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What does the solution look like when the hillside has fences?
How do we write a whole motion as one problem, and how does the solver handle the fences?

Where the optimization deck left us

- Gauss-Newton on the inverse-kinematics cost $f(\mathbf{q}) = \frac{1}{2}\|\varphi(\mathbf{q}) - \mathbf{y}^*\|^2$
- Target $\mathbf{y}^* = (0.5, 0.2)$: both answers bend the elbow 80.4°
- Add a 60° elbow stop and clip q_2 after each step:
  - the run stalls at $\mathbf{q} = (-16.4^\circ, 60^\circ)$
  - the tip is 14.6 cm off
  - the cost still falls as q_1 grows, the joint the stop leaves free
- "As close as possible" is not **"allowed"**: joint ranges, obstacles, goals that hold *exactly*
- A soft penalty trades trespass for cost

New question: what does the minimum look like with fences?

The nonlinear program, and the robotics dictionary

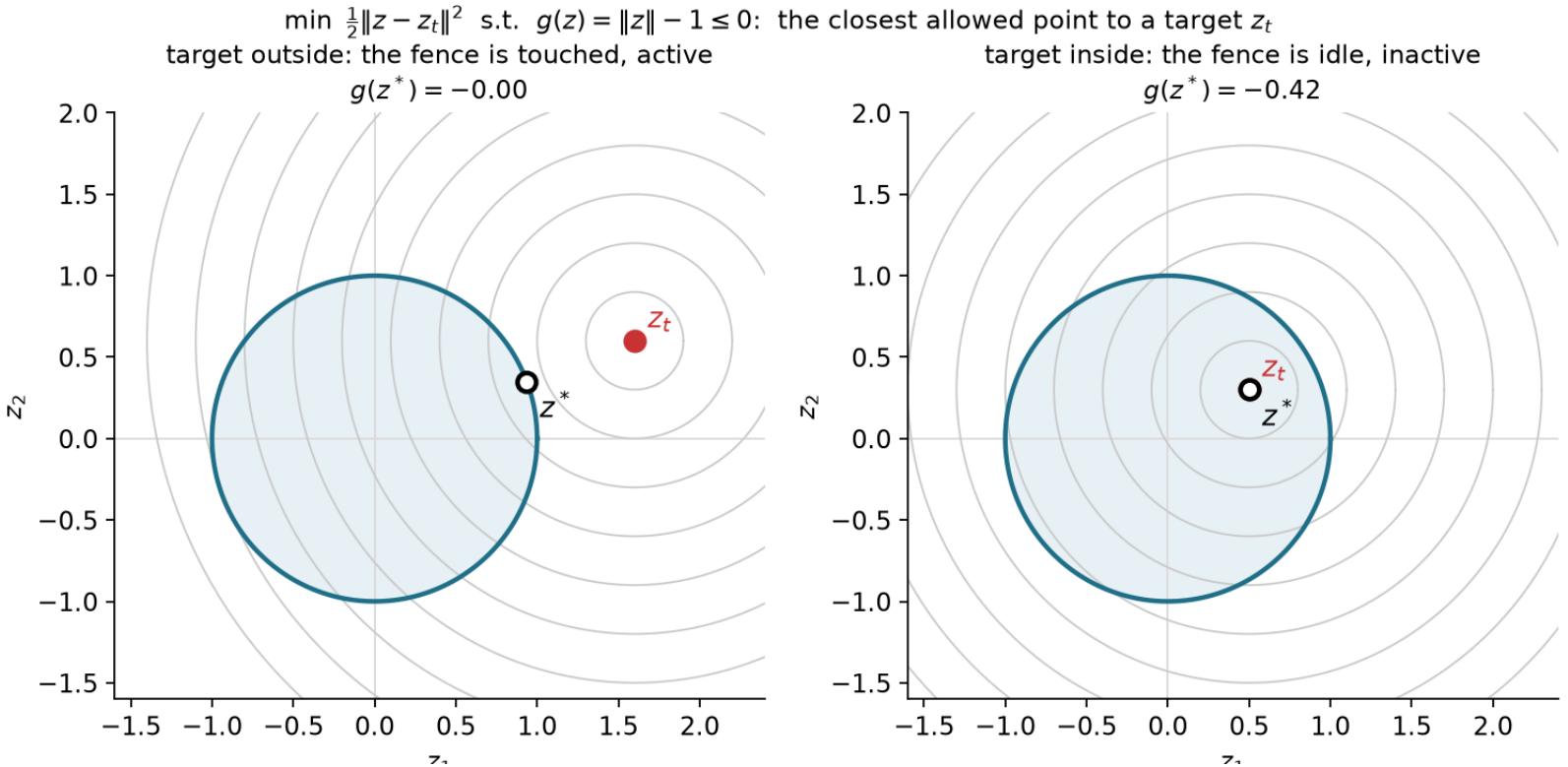
$$\min_{\mathbf{z}\in\mathbb{R}^d} f(\mathbf{z}) \quad \text{s.t.} \quad g_i(\mathbf{z}) \le 0 \;\; (i = 1..m), \qquad h_j(\mathbf{z}) = 0 \;\; (j = 1..l)$$

requirement	as a constraint	type
joint within its range	$q_j - q_j^{\max} \le 0, \quad q_j^{\min} - q_j \le 0$	inequality g
a point on the arm clear of a sphere $(\mathbf{c}, r_o)$	$r_o - \lVert\mathbf{p}(\mathbf{q}) - \mathbf{c}\rVert \le 0$	inequality g
torque bound	$\tau_j - \tau_j^{\max} \le 0, \quad -\tau_j^{\max} - \tau_j \le 0$	two inequalities g (need dynamics; not used today)
tip exactly at the goal	$\varphi(\mathbf{q}) - \mathbf{y}^* = \mathbf{0}$	equalities $\mathbf{h}$
arm at rest at the end	$\mathbf{q}_T - \mathbf{q}_{T-1} = \mathbf{0}$	equalities $\mathbf{h}$ (a trajectory: slide 27)

Convention: feasible means $g \le 0$. $g > 0$ where you must not be, so ∇g points to the forbidden side.

fence	g	∇g
elbow limit	$g_1 = q_2 - 60^\circ$	$\nabla g_1 = (0, 1)$
clearance	$r_o - \lVert\mathbf{p}(\mathbf{q}) - \mathbf{c}\rVert$	$-\mathbf{J}_p(\mathbf{q})^\top \hat{\mathbf{n}}$; $\hat{\mathbf{n}}$: unit vector from $\mathbf{c}$ to $\mathbf{p}$

Feasible set, active and inactive constraints



$\min_{\mathbf{z}} \frac{1}{2}\|\mathbf{z} - \mathbf{z}_t\|^2$ subject to $\|\mathbf{z}\| \le 1$: the closest allowed point to a target

The fence: force balance at the solution

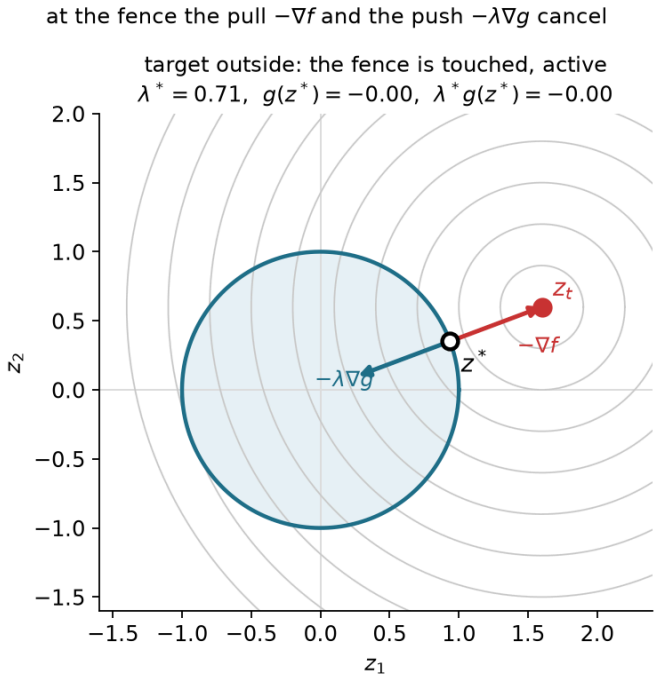
Boundary optimum $\mathbf{z}^*$, $g(\mathbf{z}^*) = 0$:

1. Deck 04's test $\nabla f = \mathbf{0}$ fails
2. No step into the allowed side descends (Deck 04 slide 6):

$$\nabla g(\mathbf{z}^*)^\top \mathbf{d} < 0 \;\Rightarrow\; \nabla f(\mathbf{z}^*)^\top \mathbf{d} \geq 0$$

3. This forces:

$$\nabla f(\mathbf{z}^*) + \lambda \nabla g(\mathbf{z}^*) = \mathbf{0}, \qquad \lambda \geq 0$$



- Pull $-\nabla f$ **downhill**; push $-\lambda \nabla g$ **normal to the fence, into the allowed side** (∇g points out of it, slide 3); they cancel
- λ = **how hard the fence pushes**, the Lagrange multiplier; never pulls, so $\lambda \geq 0$; untouched fence: $\lambda = 0$

Worked example (a): the round fence, in closed form

$$\min_{\mathbf{z}} \ \tfrac{1}{2}\|\mathbf{z}-\mathbf{z}_t\|^2 \quad \text{s.t.} \quad g(\mathbf{z}) = \|\mathbf{z}\| - 1 \le 0$$

- Inside ($\|\mathbf{z}_t\| \le 1$): $\mathbf{z}^* = \mathbf{z}_t$, $\lambda^* = 0$
- Outside: $\mathbf{z}^*$ on the circle, $(\mathbf{z}^* - \mathbf{z}_t) + \lambda \mathbf{z}^* / \|\mathbf{z}^*\| = \mathbf{0} \Rightarrow \mathbf{z}^*(1+\lambda) = \mathbf{z}_t$

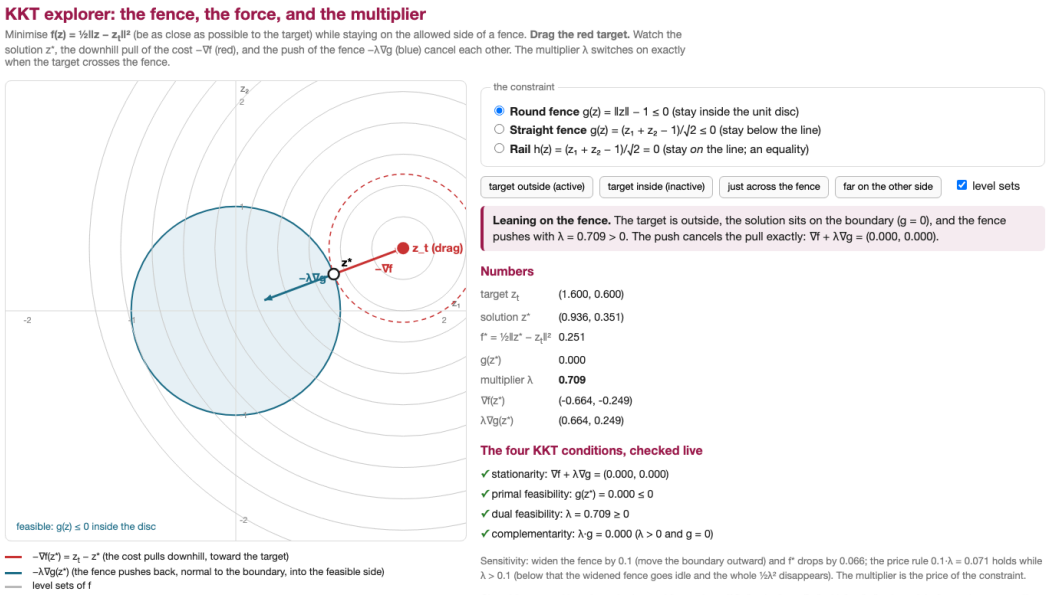
$$\mathbf{z}^* = \frac{\mathbf{z}_t}{\max(1, \|\mathbf{z}_t\|)}, \qquad \lambda^* = \max(0,\ \|\mathbf{z}_t\| - 1)$$

target $\mathbf{z}_t$	$\mathbf{z}^*$	λ^*	$g(\mathbf{z}^*)$
$(2, 0)$	$(1, 0)$	1	0
$(1.2, 0.9)$	$(0.8, 0.6)$	0.5	0
$(0.5, 0.3)$	$(0.5, 0.3)$	0	-0.42

- **This cost, unit ∇g:** λ^* = how far the target lies beyond the fence
- In general: λ^* is a price (slide 13)

Live demo: the KKT explorer

- Opens on the round fence, target inside at $(0.5, 0.3)$: $\lambda = 0$, $g = -0.417$
- **Predict:** λ for a target at $(0.9, 0.9)$
- Click **(0.9, 0.9)**: $\lambda = 0.273 = \|\mathbf{z}_t\| - 1$
- Click **target outside** $(1.6, 0.6)$: $\lambda = 0.709$, the picture
- Click **just across** $(1.02, 0)$: $\lambda = 0.020$, $g = 0$; the switch is in the activity, λ has no jump
- Arrows $-\nabla f$ and $-\lambda \nabla g$ cancel; four KKT conditions ticked live (closed form, they cannot fail)
- Next slide: the straight fence and the **rail**; slide 13: the **price readout**



modules/kkt-explorer/index.html, round fence, target inside ▶ run it live

Rails: an equality constraint holds on either side

1. Rail $h(\mathbf{z}) = 0$, approached from both sides: feasible directions $\pm\mathbf{d}$ with $\nabla h^\top \mathbf{d} = 0$
2. Neither may decrease f: $\nabla f^\top \mathbf{d} = 0$ for every $\mathbf{d} \perp \nabla h$
3. Hence $\nabla f(\mathbf{z}^*) = -\kappa\, \nabla h(\mathbf{z}^*)$, κ **of either sign**: the rail pushes from whichever side the cost pulls

- Always touched: no on/off question, unlike a fence
- Vector rail $\mathbf{h}(\mathbf{q}) = \varphi(\mathbf{q}) - \mathbf{y}^* = \mathbf{0}$, tip on the mug:
  - one scalar κ_j per row
  - $\partial\mathbf{h}/\partial\mathbf{q}$ stacks their gradients
- Module, with the disc's cost:
  - rail $h = (z_1 + z_2 - 1)/\sqrt{2}$
  - $\kappa^* = h(\mathbf{z}_t)$: signed distance to the rail

modules/kkt-explorer, straight fence ▶ run it live

The four KKT conditions

$$\nabla f(\mathbf{z}^*) + \sum_i \lambda_i \nabla g_i(\mathbf{z}^*) + \sum_j \kappa_j \nabla h_j(\mathbf{z}^*) = \mathbf{0} \quad \text{stationarity: forces balance}$$

$$g_i(\mathbf{z}^*) \le 0, \qquad h_j(\mathbf{z}^*) = 0 \quad \text{primal feasibility: on the allowed side, on every rail}$$

$$\lambda_i \ge 0 \ \ (\kappa_j\text{: any sign, previous slide}) \quad \text{dual feasibility: fences push, never pull}$$

$$\lambda_i \, g_i(\mathbf{z}^*) = 0 \quad \text{complementarity: a fence pushes only if touched}$$

- **Karush-Kuhn-Tucker (KKT)** conditions: **necessary** at a local minimum, if the touched fences' and the rails' gradients are linearly independent there (a *constraint qualification*)
- **Not a certificate** of a minimum: the straight arm $(21.8^\circ, 0^\circ)$, Deck 04 slide 18's saddle, passes all four with no fence touched (notes §5 (m))
- Complementarity: fences only; a rail is always touched
- Every solver of 4.2 aims at a point that satisfies them

Reading KKT: a weighted sum, and an on/off logic

Stationarity: a weighted sum

$$-\nabla f(\mathbf{z}^*) \in \text{cone}\{\nabla g_i\}_{\text{active}} + \text{span}\{\nabla h_j\}$$

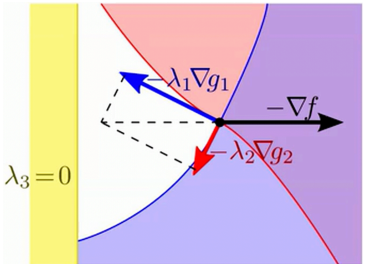
- cone: **nonnegative** weighted sums of the touched fences' gradients
- span: weighted sums **of any sign** of the rails' gradients
- each multiplier: the weight on one gradient

Complementarity: on/off, for each i:

$$\lambda_i = 0 \quad \text{or} \quad g_i(\mathbf{z}^*) = 0 \quad \text{(or both)}$$

If we *knew* which fences are touched:

1. drop the idle ones
2. turn the touched ones into equalities
3. solve one square system



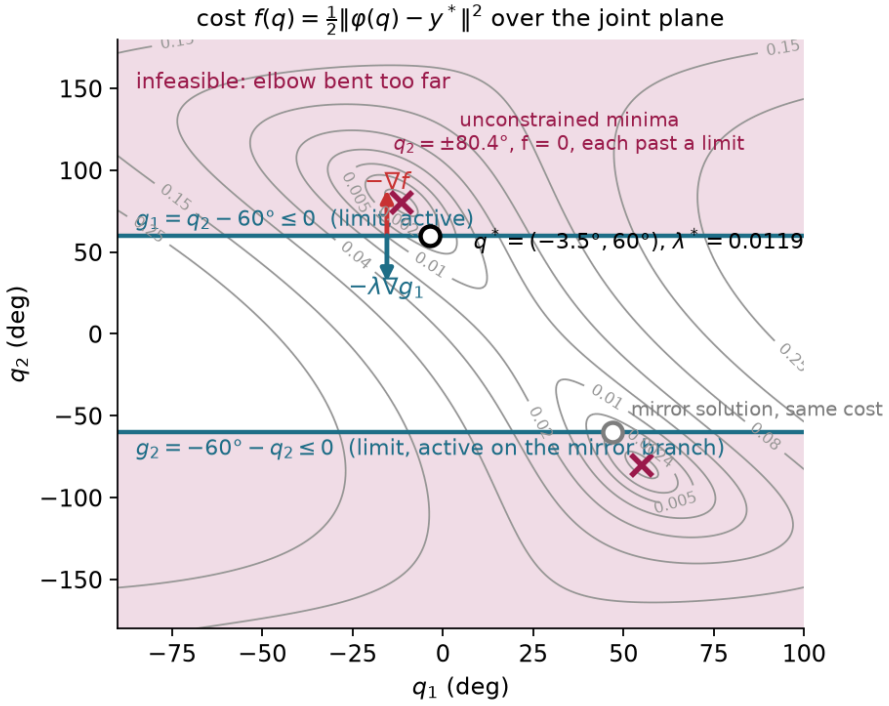
2^m **candidate active sets.** Every algorithm of 4.2 avoids enumerating them.

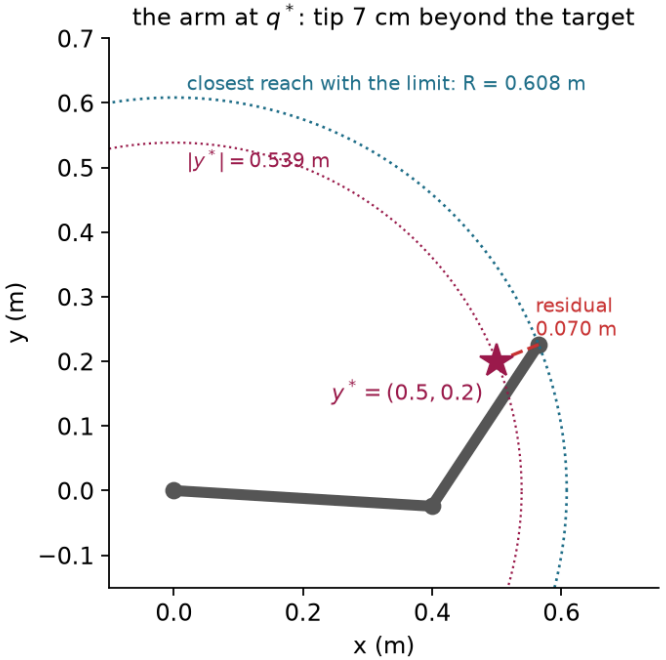
The Lagrangian, and the loop it suggests

$$L(\mathbf{z}, \boldsymbol{\lambda}, \boldsymbol{\kappa}) = f(\mathbf{z}) + \boldsymbol{\lambda}^\top \mathbf{g}(\mathbf{z}) + \boldsymbol{\kappa}^\top \mathbf{h}(\mathbf{z})$$

- **Stationarity** is exactly $\nabla_{\mathbf{z}} L(\mathbf{z}^*, \boldsymbol{\lambda}^*, \boldsymbol{\kappa}^*) = \mathbf{0}$: every constraint gradient in one force balance
- With $\boldsymbol{\lambda}^*, \boldsymbol{\kappa}^*$: $\mathbf{z}^*$ a **stationary point** of L over $\mathbf{z}$, not in general a minimizer (the elbow's L has no global minimum)
- Standard sufficient condition: upward curvature **only along the constraints**
- Multipliers unknown in advance, so the loop of slide 23:
  - **inner**: hold $\boldsymbol{\lambda}, \boldsymbol{\kappa}$ fixed, minimize over $\mathbf{z}$
  - **outer**: correct the multipliers from the violations
- An added squared term supplies curvature in the constraint-normal directions

Worked example (b): the two-link arm with an elbow limit





limit	broken by the unconstrained answer	at $\mathbf{q}^* = (-3.5^\circ, 60^\circ)$
$g_1 = q_2 - 60^\circ \le 0$	$q_2 = +80.4^\circ$	active, $\lambda_1^* = -\partial f / \partial q_2 = -\mathbf{J}_{:,2}^\top \mathbf{r} = 0.0119$
$g_2 = -60^\circ - q_2 \le 0$	$q_2 = -80.4^\circ$	idle, $\lambda_2^* = 0$

Slide 2's clamped point has $\partial f / \partial q_1 = -0.073$, so it is not a KKT point.

The multiplier is a price

Move the fence to $g(\mathbf{z}) \le b$; optimal cost $f^*(b)$.

1. Differentiate $g(\mathbf{z}^*(b)) = b$:
   $\nabla g^\top d\mathbf{z}^*/db = 1$
2. Chain rule, then stationarity:

$$\frac{df^*}{db} = \nabla f^\top \frac{d\mathbf{z}^*}{db} = -\lambda \nabla g^\top \frac{d\mathbf{z}^*}{db} = -\lambda$$

Locally, λ = cost saved per unit of relaxation

Check on the arm: relax the elbow limit by 1°.

quantity	value	unit
λ^*	0.01192	m²/rad
1°	0.01745	rad
predicted change, $-\lambda^* \cdot 0.01745$	-0.000208	m²
computed f^* before	0.002433	m²
computed f^* after	0.002228	m²
computed change	-0.000205	m²

- Valid on one branch, active set unchanged
- One-sided here: tightening selects the mirror branch (notes §4.4)
- Inactive fence: $\lambda = 0$, moving it changes nothing

modules/kkt-explorer, the price line ▶ run it live

Where constraints enter the stack

stack box	where constraints enter
Goal (4.2, solved first)	inverse kinematics with an exact target, joint ranges and clearance
Reference motion (4.2)	a whole trajectory, the same constraints at every time step: KOMO; sampling-based alternative in 4.4
Controller (4.3)	differential inverse kinematics (IK) as a *quadratic program* with velocity limits: this KKT machinery a thousand times per second
Learned counterparts (5.1, 7.1)	a plain policy output: no feasibility certificate, no multiplier; a learned proposal can still feed a constrained solver, where the two meet

- Planned for 4.3: the essentials' two deferred items, constrained differential IK and secondary motion that leaves the task unchanged

Summary of 4.1

- Constraints: fences $g \leq 0$ and rails $h = 0$; feasible means $g \leq 0$
- At a fenced optimum the downhill pull and the fence's push cancel: $\nabla f + \lambda \nabla g = \mathbf{0}$, $\lambda \geq 0$
- Complementarity $\lambda g = 0$: a fence pushes only if touched; 2^m patterns is the difficulty
- λ is a price: $df^*/db = -\lambda$

Break

Second half: inverse kinematics as one nonlinear program, three ways a solver builds a fence out of costs, and whole trajectories solved the same way.

Inverse kinematics as a nonlinear program

$$\min_{\mathbf{q}} \ \tfrac{1}{2}\|\mathbf{q}-\mathbf{q}_{\mathrm{ref}}\|_{\mathbf{D}}^{2} \quad \text{s.t.} \quad \varphi(\mathbf{q})-\mathbf{y}^{*}=\mathbf{0}, \quad \mathbf{q}^{\min}\le\mathbf{q}\le\mathbf{q}^{\max}, \quad r_o-\|\mathbf{p}_p(\mathbf{q})-\mathbf{c}\|\le 0$$

- Two changes from Deck 04:
  - target: soft residual → **hard equality**
  - cost: a **fixed** reference posture $\mathbf{q}_{\mathrm{ref}}$ replaces the step penalty
- $\mathbf{q}_{\mathrm{ref}}$ elsewhere: $\mathbf{q}_0$ in the essentials' §5, $\mathbf{q}_{\mathrm{ready}}$ in the Panda demo
- With the elbow limited to $\pm 60^\circ$:
  - reach: 0.608 to 0.700 m from the base
  - target: 0.539 m away, so an exact target is **infeasible**
- Same slot, more features: gaze, feet stay put, balance, contacts (Deck 06, A2)

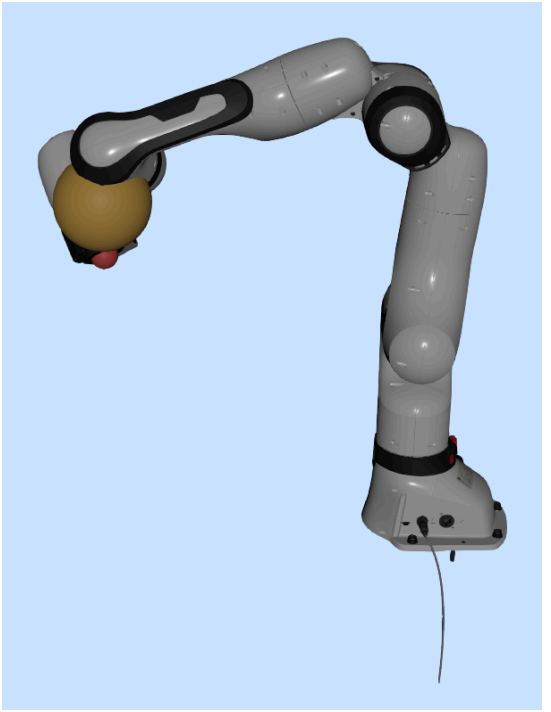
Spring 2026 deck 03, motion planning, p.53 and p.60

Live demo: IK as a nonlinear program on the Panda

Slide 17's program on a 7-joint arm; one solver, SNOPT, through Drake's `InverseKinematics`

term	on the Panda
cost	$\frac{1}{2}\|\mathbf{q} - \mathbf{q}_{\text{ready}}\|^2$
rail	flange on the target: $\varphi(\mathbf{q}) - \mathbf{y}^* = \mathbf{0}$
fences	URDF joint limits; three wrist points clear of a sphere $(\mathbf{c}, r_o)$

live start	value
target $\mathbf{y}^*$	$(0.45, 0.10, 0.45)$ m
sphere center $\mathbf{c}$	$(0.45, 0.10, 0.53)$ m, 8 cm above the target
radius r_o	0.07 m



- **Predict:** the printed active set; its change 10 and 15 cm lower; a target inside the sphere
- Live: three sliders; the console prints cost and active set; a failed solve keeps the arm, greys the target

drake/03-ik-as-nlp/ik_nlp_panda.py --meshcat ▶ run it live

Panda results: active sets, multipliers and prices

run	target $\mathbf{y}^*$ (m)	active inequalities	λ	$\boldsymbol{\kappa}$ (target x, y, z)	cost
1	$(0.45, 0.10, 0.45)$, no sphere	none	$-$	$(-2.07, -0.67, 0.71)$	0.209
2	$(-0.20, 0, 0.60)$, behind the robot	joint 2 at its lower limit -1.7628 rad	4.196	$(8.20, 0, 14.93)$	1.729
3	run 1 + the sphere (the live start)	wrist point at 0.0700 m from $\mathbf{c}$	7.30	$(2.63, -5.76, -0.02)$	0.386

price check	shift	predicted	computed
run 1, target x	$+1$ mm	$-\kappa_x \times 0.001 = +0.00207$	$+0.00208$
run 2, joint-2 limit	$+0.001$ rad	$\lambda \times 0.001 = +0.00420$	$+0.00422$

- The rail is imposed in every run; its $\boldsymbol{\kappa}$ is nonzero here, but need not be
- Runs 1 and 2 come from print mode; run 2's target lies outside the slider range
- Console "no feasible point found": a solver failure, **not** a proof; inside the sphere it is infeasible by construction

drake/03-ik-as-nlp/ik_nlp_panda.py (print mode) · notes §5 (k)

How does a solver handle the fences? Three ways to build one out of costs

construction	augmented objective	outer loop	character
squared penalty	$S = f + \mu \sum_i [g_i > 0]\, g_i^2 + \mu \sum_j h_j^2$	$\mu \to \infty$	a ramp: trespasses wherever a fence must push ($\lambda^* > 0$)
log barrier	$B = f - \mu \sum_i \log(-g_i)$	$\mu \to 0$	a wall inside the fences: never touches, needs a feasible start; encodes fences only
augmented Lagrangian	$A = f + \sum_i [g_i > 0 \vee \lambda_i > 0](\lambda_i g_i + \mu g_i^2) + \sum_j (\kappa_j h_j + \mu h_j^2)$	$\boldsymbol{\lambda}, \boldsymbol{\kappa}$ updated between inner solves, μ fixed	a ramp plus a learned wind

Each inner problem: **unconstrained**, solved by Deck 04's Newton or Gauss-Newton (quasi-Newton BFGS: Deck 04's backup slides).

Squared penalty: a ramp on the forbidden side

$$S(\mathbf{z};\mu) = f(\mathbf{z}) + \mu\,[g > 0]\,g(\mathbf{z})^2$$

1. Inner minimum: $\nabla f + 2\mu g\,\nabla g = \mathbf{0}$
2. So $2\mu g$ plays λ, and the violation is

$$g(\mathbf{z}') \approx \frac{\lambda^*}{2\mu}$$

- **Trespass wherever the fence must push** ($\lambda^* > 0$), shrinking like $1/\mu$
- An idle fence is not violated
- Hessian gains an eigenvalue of order μ: conditioning worsens as the answer improves

Elbow limit, from $(30^\circ, 45^\circ)$:

μ	q_2 (deg)	violation	$2\mu g$
1	60.338	0.338°	0.01181
10	60.034	0.034°	0.01191
100	60.003	0.003°	0.01192
1000	60.000	0.0003°	0.01192

$\lambda^* = 0.01192$

μ is chosen; λ is solved for (by hand on slides 6 and 12, by the outer loop on slide 23)

Essentials §5: the posture-weighted cost with a fixed $\mathbf{q}_{\text{ref}}$ is this penalty on the target rail, $\mu = K/2$

Log barrier: a wall that is never touched

$$B(\mathbf{z};\mu) = f(\mathbf{z}) - \mu \log(-g(\mathbf{z}))$$

$$(1)\ \text{stationarity of } B: \quad \nabla f - \tfrac{\mu}{g}\nabla g = \mathbf{0}$$

$$(2)\ \text{name the push:} \quad \lambda := -\mu/g > 0$$

$$(3)\ \text{substitute:} \quad \nabla f + \lambda \nabla g = \mathbf{0}, \qquad \lambda\, g = -\mu$$

Complementarity relaxed by μ

Limits:

- strictly feasible start only
- added cost encodes fences only: a rail needs an equality-constrained inner solve (interior-point codes do that)
- Hessian term $\lambda^2/\mu\, \nabla g \nabla g^\top$ grows as $\mu \to 0$ (active fence)

Elbow limit, from $(30^\circ, 45^\circ)$:

μ	q_2 (deg)	slack	$-\mu/g$
10^{-2}	28.833	31.2°	0.01838
10^{-3}	55.786	4.2°	0.01360
10^{-4}	59.527	0.47°	0.01212
10^{-6}	59.995	0.005°	0.01192

- every iterate strictly inside
- **Central path:** the minima $\mathbf{z}'(\mu)$ trace it to the optimum as $\mu \to 0$

Augmented Lagrangian: a ramp plus a learned wind

$$A = f + \lambda g + \mu g^2 \quad (\text{while } g > 0 \text{ or } \lambda > 0)$$

1. **Inner**, multipliers frozen: minimize A over $\mathbf{z}$; at its minimum:

$$0 = \nabla f + (\lambda + 2\mu g)\,\nabla g$$

the parenthesis: the new estimate of λ^*

2. **Outer**, $\mathbf{z}'$ frozen: update the multipliers

$$\lambda \leftarrow \max(\lambda + 2\mu g(\mathbf{z}'),\ 0), \qquad \kappa \leftarrow \kappa + 2\mu h(\mathbf{z}')$$

The inner problem is least squares

Complete the square; the dropped constant does not depend on $\mathbf{z}$:

$$\lambda g + \mu g^2 = \tfrac{1}{2}\left(\sqrt{2\mu}\,(g + \tfrac{\lambda}{2\mu})\right)^2 - \tfrac{\lambda^2}{4\mu}$$

Inner cost: $\tfrac{1}{2}\|\mathbf{r}\|^2$ plus a constant, with rows

residual row	value	Jacobian row
fence with $g > 0$ or $\lambda > 0$	$\sqrt{2\mu}\,(g + \frac{\lambda}{2\mu})$	$\sqrt{2\mu}\,\nabla g^\top$
rail	$\sqrt{2\mu}\,(h + \frac{\kappa}{2\mu})$	$\sqrt{2\mu}\,\nabla h^\top$

so **Deck 04's damped step solves it** (λ, κ fixed)

Toussaint 2014, arXiv:1412.4329

The augmented Lagrangian learns the multiplier

Theory: a fixed μ suffices

- above a problem-dependent threshold
- a local result

	μ	stop
our code	fixed	three residuals (slide 25)
RAI's `NLP_Solver` default (documented; not executed here)	$\times 5$ per outer step	step tolerance 0.01

Elbow limit, $\mu = 1$ fixed:

outer	λ before	q_2 (deg)	g	λ after
0	0	60.338	$+0.338^\circ$	0.011805
1	0.011805	60.003	$+0.003^\circ$	0.011917
2	0.011917	60.000	$+0.00003^\circ$	0.011918
3	0.011918	60.000	0	0.011918

lower limit: $\lambda_2 = 0$ throughout

Toussaint 2014, arXiv:1412.4329 · modules/opt-explorer, Constrained tab ▶ run it live

Stopping with fences: three residuals, not one gradient

residual (updated multipliers)	meaning	elbow limit, AL $\mu = 1$, outer 3	KOMO solution A, outer 5
feasibility: $\max g^{+}$	on the allowed side	4.9×10^{-9}	0
feasibility: $\max\lvert h_j \rvert$	on every rail	— (no rail)	5.1×10^{-10}
Lagrangian stationarity: $\lVert \nabla f + \sum_i \lambda_i \nabla g_i + \sum_j \kappa_j \nabla h_j \rVert$	forces balance	1.3×10^{-11}	1.3×10^{-10}
complementarity: $\max_i \lvert \lambda_i g_i \rvert$	idle fences push nothing	5.9×10^{-11}	9.5×10^{-13}

$\lVert \nabla f \rVert$ alone: **not a sufficient test**

- elbow optimum: $\lVert \nabla f \rVert = 0.0119$
- KOMO path A: $\lVert \nabla f \rVert = 0.027$, stays there
- another constrained optimum: can be 0

Deck 04's words, new meanings:

status	meaning
solved	all three residuals small
stationary	force balance holds (current multipliers); another residual not settled
stalled	an inner solve rejects its direction
limit	outer iteration cap reached

code/constrained_ik.py (d) · code/komo_two_link.py · code/check_solvers.py

Comparison

	squared penalty	log barrier	augmented Lagrangian
added cost encodes rails	yes	**no**: a rail needs an equality-constrained inner solve	yes
start anywhere	yes	**no**: strictly feasible start	yes
iterates feasible	no: trespass where $\lambda^* > 0$	yes, interior	no, until the multipliers settle
exact in the limit	$\mu \to \infty$	$\mu \to 0$	fixed μ above a threshold, as the multipliers converge (local)
conditioning	worsens as $\mu \to \infty$	worsens as $\mu \to 0$	μ need not grow for feasibility; the problem's own conditioning remains
where you meet it	soft costs everywhere	interior-point solvers (IPOPT)	KOMO; RAI's `NLP_Solver` (documented default $\mu \times 5$ per outer step; not run here)

From one configuration to a path

- Knots $\mathbf{q}_0, \mathbf{q}_1, \ldots, \mathbf{q}_T$; $\mathbf{q}_0$ fixed: the current configuration
- Costs on **differences**, today $\Delta t = 1$:
  - velocity $(\mathbf{q}_t - \mathbf{q}_{t-1})/\Delta t$
  - acceleration $(\mathbf{q}_{t-2} - 2\mathbf{q}_{t-1} + \mathbf{q}_t)/\Delta t^2$
- Constraints **at time slices**:
  - goal and rest: $t = T$
  - clearance: every t

One nonlinear program, nT variables.

$$\min_{\mathbf{q}_{1:T}} \sum_t \tfrac{1}{2} \left\| \mathbf{r}_t(\mathbf{q}_{t-k:t}) \right\|^2$$

$$\text{s.t. } \mathbf{g}_t(\mathbf{q}_{t-k:t}) \le \mathbf{0}, \;\; \mathbf{h}_t(\mathbf{q}_{t-k:t}) = \mathbf{0}$$

decision variable:

$$\mathbf{z} = (\mathbf{q}_1, \ldots, \mathbf{q}_T) \in \mathbb{R}^{nT}$$

$T = 1$, no rest constraint: slide 17's inverse kinematics, with

- $\mathbf{q}_{\text{ref}} = \mathbf{q}_0$
- $\mathbf{D} = \mathbf{I}$

Spring 2026 deck 03, motion planning, p.61; Toussaint 2017

The k-order Markov structure (KOMO)

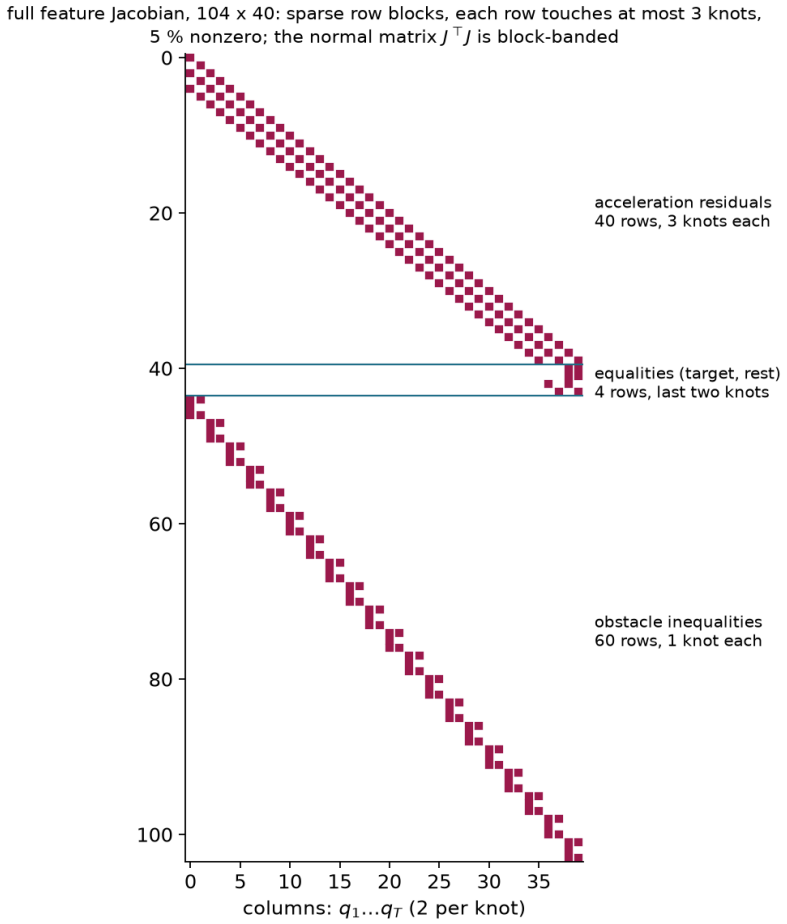
- A feature uses at most $k+1$ consecutive knots; k is the **Markov order**

k	features
0	position, distance, orientation
1	velocity
2	acceleration

$$\mathbf{r}_t = \mathbf{q}_{t-2} - 2\mathbf{q}_{t-1} + \mathbf{q}_t, \quad t = 1..T$$
$$\mathbf{h} = \big(\varphi(\mathbf{q}_T) - \mathbf{y}^*,\ \mathbf{q}_T - \mathbf{q}_{T-1}\big)$$
$$g_{t,p} = r_o - \|\mathbf{p}_p(\mathbf{q}_t) - \mathbf{c}\|$$

- Start from rest: $\mathbf{q}_{-1} := \mathbf{q}_0$, so $\mathbf{r}_1 = \mathbf{q}_1 - \mathbf{q}_0$
- A velocity cost jumps to full speed in one step; $k = 2$ penalizes that jump
- **k-order Markov Optimization (KOMO)** names this formulation; RAI's `kOrder=2` selects this order

Why k-order matters: every row is local in time



Left: full feature Jacobian

- size 104×40
- $5.3\,\%$ nonzero

Rows:

- each touches at most $k+1$ nearby knots
- goal rows' block: Deck 03's $\mathbf{J}(\mathbf{q}_T)$
- so $\mathbf{J}_r^\top \mathbf{J}_r$ is **block-banded**

Operation count of one step (an estimate, not a timing):

$$O\big(T\,(k+1)^2 n^3\big) \;\text{ instead of }\; O\big((nT)^3\big)$$

T	k	n	**banded**	**dense**
100	2	7	$\approx 3 \times 10^5$	$\approx 3 \times 10^8$

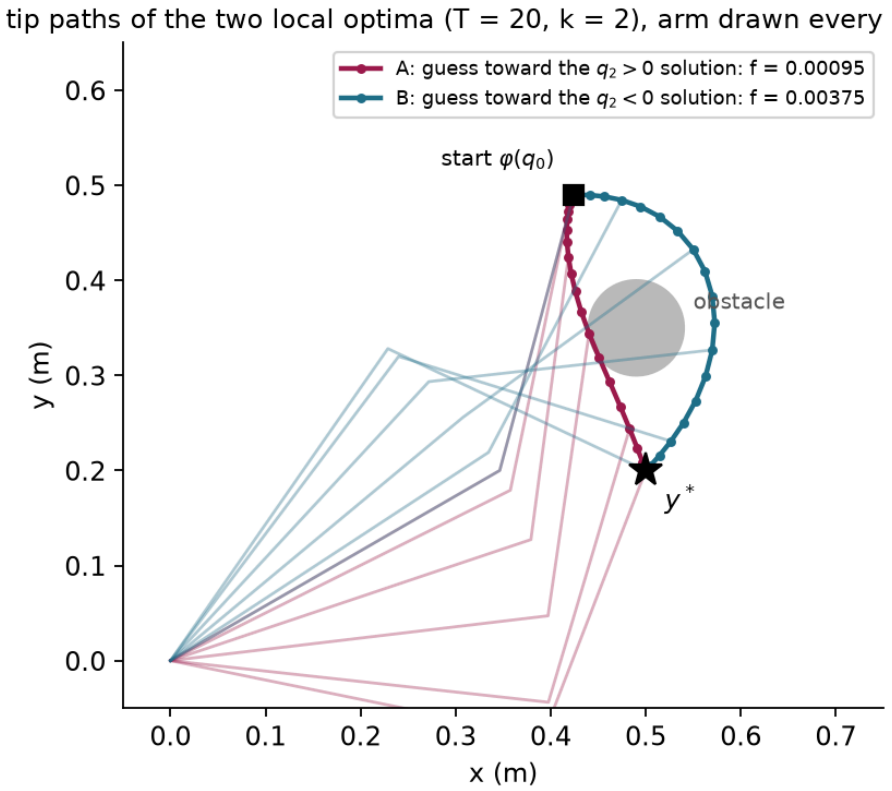
Two Jacobians, both sparse

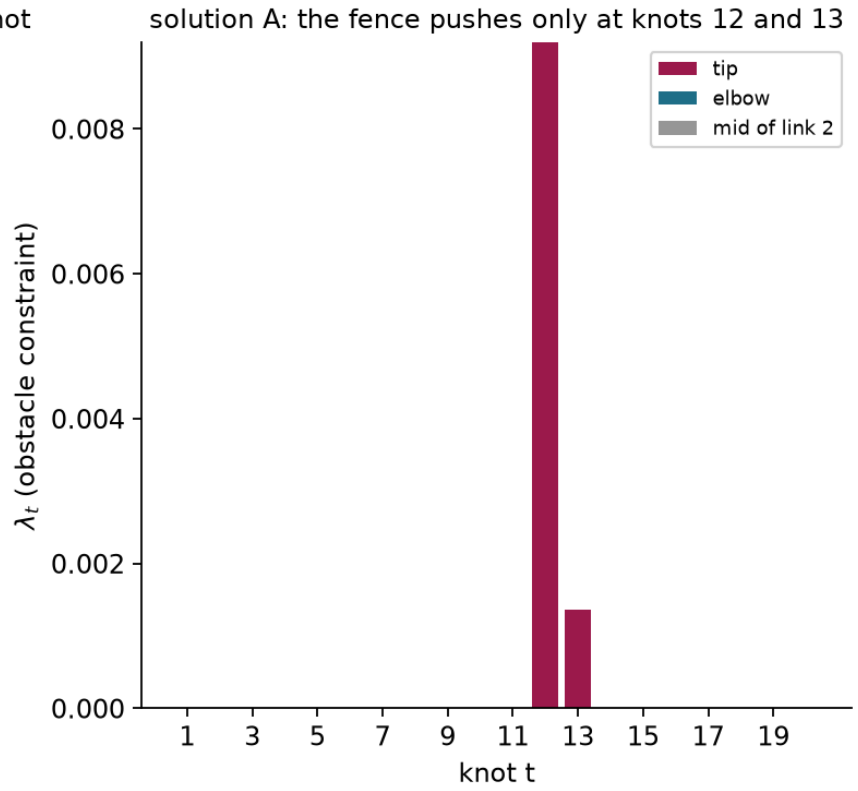
- **Full feature Jacobian:** every feature row, unscaled (slide 29's figure)
- **Inner residual Jacobian:** the rows that slide 23's inner solve factorizes

rows	full feature Jacobian	inner residual Jacobian, solution A
smoothness $\mathbf{r}_t$	40	40
rails $\mathbf{h}$	4	4, times $\sqrt{2\mu}$
fences $\mathbf{g}$	60	2 (idle ones dropped), times $\sqrt{2\mu}$
size	104×40	46×40
nonzero	$5.3\,\%$	–

- Smoothness rows alone, $T = 20$: condition number 2.5×10^5
  - grows like T^4 (Deck 04 slide 9)

Worked example (i): KOMO on the two-link arm





T	k	start $\mathbf{q}_0$	target $\mathbf{y}^*$	disc r_o	points per knot	solver	μ
20	2	$(30^\circ, 45^\circ)$	$(0.5, 0.2)$	5 cm	3	slide 23's augmented Lagrangian	10, fixed

Right: λ_t, the fence's push, only at knots 12–13. No elbow limit here: under $\pm 60^\circ$ the target is out of reach (slide 12)

Two guesses, two valleys

	guess A	guess B
initial guess: joint-space line to	the elbow $+80.4^\circ$ solution	the elbow -80.4° solution
trespass before iteration 1	4.8 cm	none
ours (augmented Lagrangian): path	**below** the disc	**above** the disc
ours: f	0.000946	0.003746
ours: contact	knots 12–13	none
Drake + SNOPT: f	0.000957, a *different* KKT point	same path to 0.003°, same f
Drake + SNOPT: contact	knots 11–12	none
Drake + SNOPT: KKT residual	about 4×10^{-7}	—

- Ours, restarted at SNOPT's point A: does not move
- Drake's rail duals $= -\boldsymbol{\kappa}$ (Drake's sign)

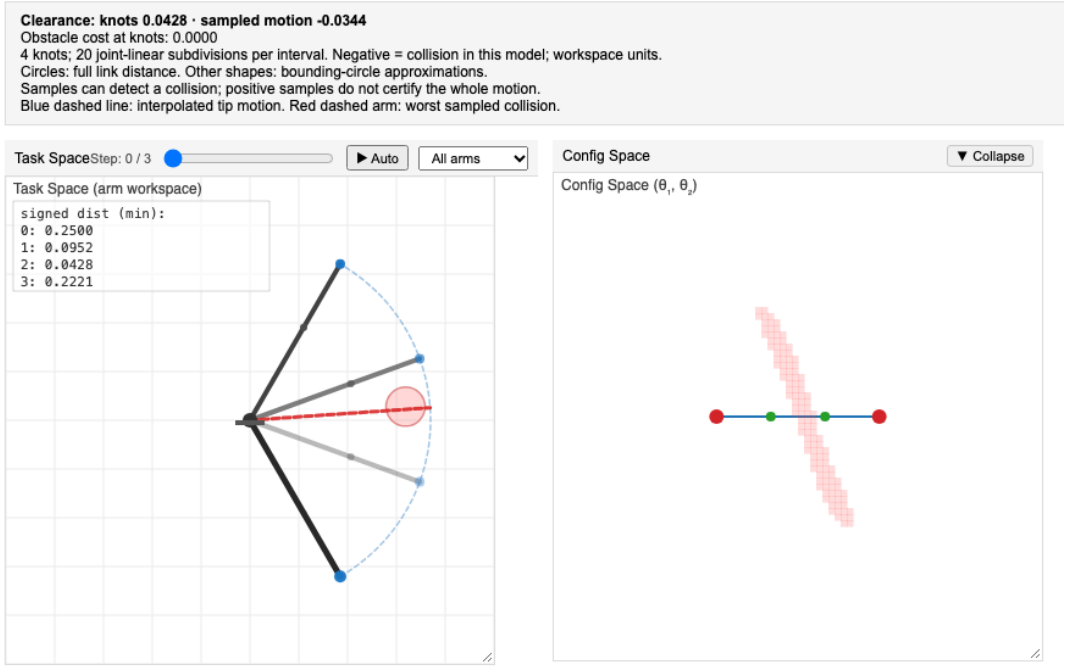
Fog: the guess decides the valley. **Report the guess with the answer.**

code/komo_two_link.py --drake

Live demo: clear knots can hide a collision

- Opens on the **Trajectory** tab with **Between-knot collision** loaded: four arm poses, all clear; obstacle cost **0**
- **Predict:** is the motion between those poses clear too?
- Interpolate the joints: the **red dashed arm collides**
  - knot clearance: positive
  - sampled-motion clearance: negative
- Choose **Add midpoint knots**: same joint path, seven knots, now a **positive obstacle cost**

More knots reveal a missed collision. They do not repair the path or certify continuous clearance.



modules/opt-explorer · Trajectory tab · Between-knot collision **▶ run it live**

KOMO as a Drake program, line by line

```
# code/komo_two_link.py --drake: slide 28's program, handed to SNOPT
prog = MathematicalProgram()
z = prog.NewContinuousVariables(N, "z")                          # z = (q_1, …, q_T)
_, A = acc_residuals(np.zeros(N)); b, _ = acc_residuals(np.zeros(N))  # r = A z + b
prog.Add2NormSquaredCost(np.sqrt(0.5)*A, -np.sqrt(0.5)*b, z)  # cost ½‖A z + b‖²
bh = prog.AddConstraint(h_fn, np.zeros(4), np.zeros(4), z)    # rails
bg = prog.AddConstraint(g_fn, -np.inf*np.ones(3*T), np.zeros(3*T), z)   # fences
prog.SetInitialGuess(z, z_init); result = SnoptSolver().Solve(prog)
zd = result.GetSolution(z); lam = result.GetDualSolution(bg)  # path, fence duals
```

Slide 28's three kinds of term, nothing else:

term	Drake call	`lb`	`ub`
cost	`Add2NormSquaredCost` (an `AddCost`)	—	—
rail	`AddConstraint`	`0`	`0`
fence	`AddConstraint`	`−∞`	`0`

- Guess A: SNOPT ends at a **second KKT point** of the same program (costs on slide 32)
  - knots up to 2.9° apart from ours
  - slide 23's loop restarted there stays within 10^{-4} degrees
- Guess B: both solvers agree to 0.003°

code/komo_two_link.py --drake (how-to in the Drake README) · RAI's KOMO calls and the solver-comparison checklist: notes §6 (documentation only)

Failure modes (1): posing and stopping

- **Sign of g:** "distance at least r_o" is $r_o - \|\mathbf{p}_p - \mathbf{c}\| \leq 0$; reversed, the obstacle's inside is feasible
- **Negative λ:** fails dual feasibility, so not a KKT point as posed; a negative κ is fine
- **Penalty "converged" but violated:** a finite μ trespasses wherever the fence must push; check g, not only the step norm
- **Stopping on $\|\nabla f\|$ with fences:** 0.012 at the elbow optimum; use slide 25's three residuals

Failure modes (2): solvers and certificates

- **Barrier from outside, or on a rail:** undefined; too large a μ leaves Newton at a saddle point
- **One solver's answer is not "the optimum":** two KKT points $1\,\%$ apart from one guess; report the guess and the residuals
- **"Infeasible" from a solver:** a local failure to find a feasible point, not a proof; the elbow case is proved by reach (slide 17)
- **Constraints hold at the knots only:** 2.07 mm trespass between knots 12–13 at $T = 20$, joint-space interpolation; more knots reveal it, none certify clearance

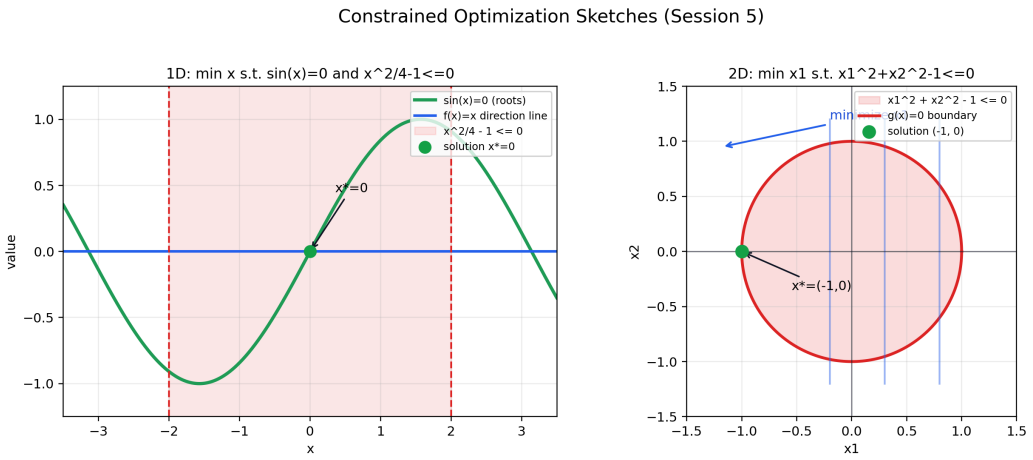
Summary; quiz 2; A1; the next deck

- **4.1:** fences and rails; force balance $\nabla f + \lambda \nabla g = \mathbf{0}$; a fence pushes only if touched (complementarity); λ is a price
- **4.2:** inverse kinematics and trajectories as one nonlinear program
  - k-order features: Jacobian rows local in time, so $\mathbf{J}^\top \mathbf{J}$ is block-banded
  - fences into costs: penalty, barrier, augmented Lagrangian (RAI's documented default; not run here)
  - done when feasibility, force balance and complementarity are all small
- **R 5:** quiz 2 (Gauss-Newton, KKT, KOMO, control stack, RRT); **A1 due**
- **Deck 06 (4.3–4.4):** the control stack, PD and MPC in one slide; configuration space and sampling-based planning

Backup slides

Backup · Solve by sketching

1. $\min x$ s.t. $\sin x = 0$, $x^2/4 - 1 \leq 0$: which constraints are active, which multipliers nonzero?
2. $\min x_1$ s.t. $x_1^2 + x_2^2 - 1 \leq 0$: force balance at the solution?
3. Wedge: $\min x + y$ s.t. $0.5x - y - 0.2 \leq 0$, $-x + 0.5y - 0.2 \leq 0$
   - solution: $(x, y) = (-0.4, -0.4)$, both active
   - multipliers: $\lambda_0 = \lambda_1 = 2$



Backup · The price of a constraint, derived

Setup, perturbed problem: $\min f(\mathbf{z})$ s.t. $g(\mathbf{z}) \le b$, active near $b = 0$

- solution $\mathbf{z}^*(b)$, optimal cost $f^*(b) = f(\mathbf{z}^*(b))$
- assumption: the same fences stay active for small b, so $\mathbf{z}^*(b)$ moves smoothly

1. Differentiate the identity $g(\mathbf{z}^*(b)) = b$: $\quad \nabla g(\mathbf{z}^*)^\top \dfrac{d\mathbf{z}^*}{db} = 1$
2. Differentiate the cost; use stationarity $\nabla f = -\lambda \nabla g$:

$$\frac{df^*}{db} = \nabla f(\mathbf{z}^*)^\top \frac{d\mathbf{z}^*}{db} = -\lambda \, \nabla g(\mathbf{z}^*)^\top \frac{d\mathbf{z}^*}{db} = -\lambda$$

3. Several constraints: $\partial f^* / \partial b_i = -\lambda_i$; inactive ones have $\lambda_i = 0$ and no price

Backup · The central path, and the barrier's relaxed KKT

At the inner optimum of $B(\mathbf{z};\mu)$, with

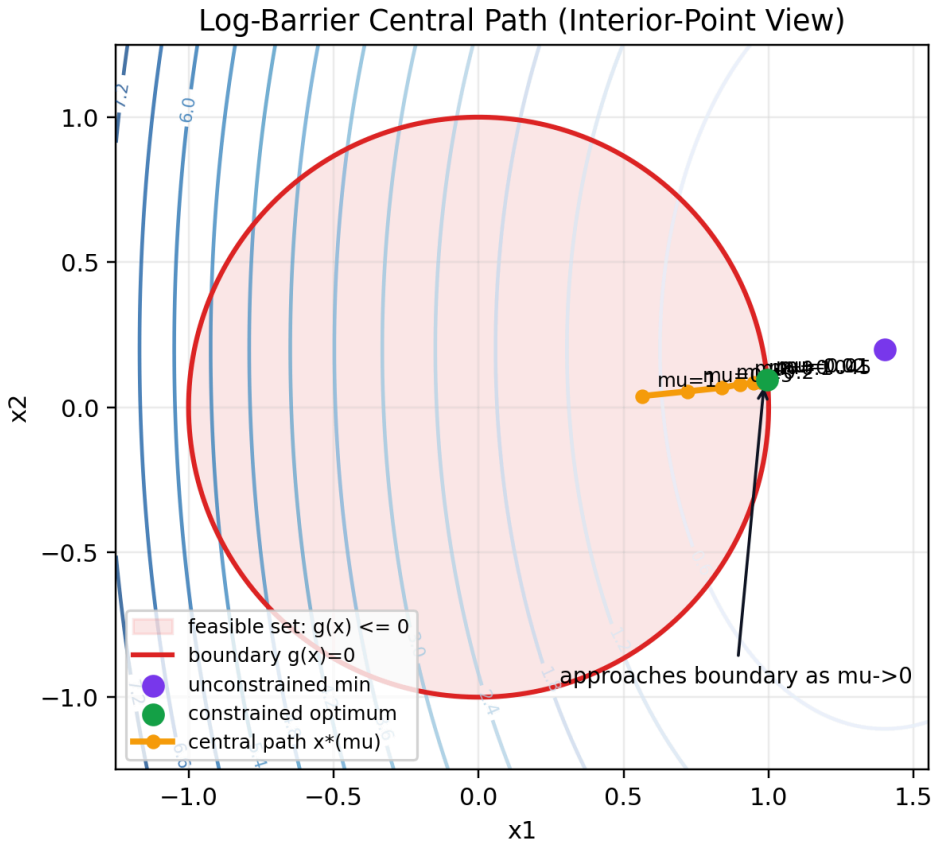
$$\lambda_i := -\mu/g_i > 0$$

condition	exact KKT	barrier optimum
stationarity	$\nabla f + \sum \lambda_i \nabla g_i = \mathbf{0}$	same
dual feasibility	$\lambda_i \geq 0$	automatic
primal feasibility	$g_i \leq 0$	strict, $g_i < 0$
complementarity	$\lambda_i g_i = 0$	$\lambda_i g_i = -\mu$

Duality gap bound (**convex** only, m fences):

$$f(\mathbf{z}'(\mu)) - f^* \leq m\mu$$

- arm's saddle run (notes §7 (6)): fails by four orders of magnitude



Backup · Gradient and Hessian of the barrier

$$\nabla_{\mathbf{z}} B = \nabla f - \mu \sum_i \frac{1}{g_i} \nabla g_i, \qquad \nabla^2_{\mathbf{z}} B \approx \nabla^2 f + \mu \sum_i \frac{1}{g_i^2} \nabla g_i \nabla g_i^\top$$

- $-\mu / g_i > 0$: a repulsive force, growing as the fence is approached
- Hessian model drops $\nabla^2 g_i$, as Gauss-Newton drops the residuals' second derivatives
- Added term: positive semidefinite, size λ_i^2 / μ
  - active fence: unbounded as $\mu \to 0$; conditioning worsens, like the penalty's
- Inner solver: Newton on B with this model, Armijo backtracking, warm start
  - stop when $m\mu$ is below the tolerance: a valid test for convex problems only

Backup · The augmented Lagrangian, derived

1. One rail: $A = f + \kappa h + \mu h^2$; stationarity at the inner minimum z':

$$0 = \nabla f + \kappa \nabla h + 2\mu h \nabla h = \nabla f + (\kappa + 2\mu h)\nabla h$$

2. Match with KKT, $\nabla f + \kappa^* \nabla h = 0$: the parenthesis is the new κ
3. Linear f and h: exact after one update

One-dimensional example (table at right):

- problem: $\min \frac{1}{2} z^2$ s.t. $z - 1 = 0$
- penalty weight: $\mu = 1$
- inner minimum: $z' = (2 - \kappa)/3$

outer	κ	z'	$h(z')$	κ new
0	0	0.6667	-0.3333	-0.6667
1	-0.6667	0.8889	-0.1111	-0.8889
2	-0.8889	0.9630	-0.0370	-0.9630
3	-0.9630	0.9877	-0.0123	-0.9877
∞	-1	1	0	-1

h shrinks by $1/(1 + 2\mu) = 1/3$ per outer iteration at **fixed** μ.

Backup · The full algorithm with inequalities

```
μ ← μ0;  λ ← 0;  κ ← 0
repeat
    z ← argmin_z  f(z) + Σ_i [g_i(z) > 0 ∨ λ_i > 0] (λ_i g_i(z) + μ g_i(z)²) + Σ_j (κ_j h_j(z) + μ h_j(z)²)     # inner: Newton / Gauss-Newton / BFGS, warm-started
    λ_i ← max(λ_i + 2μ g_i(z), 0)                                                                              # fences: clamp at zero
    κ_j ← κ_j + 2μ h_j(z)                                                                                      # rails
    optionally μ ← c μ                    # RAI's documented default: c = 5
until  max_i g_i(z)⁺, max_j |h_j(z)| ≤ 10⁻⁹          # feasibility          (slide 25's three residuals,
  and  ‖∇f + Σ_i λ_i ∇g_i + Σ_j κ_j ∇h_j‖ ≤ 10⁻⁸       # Lagrangian stationarity   with the updated multipliers;
  and  max_i |λ_i g_i(z)| ≤ 10⁻¹¹                     # complementarity       stop also if an inner solve stalls)
```

- The bracket $[g_i > 0 \vee \lambda_i > 0]$ decides activity **pointwise**, at every evaluation: no enumeration of active sets
- $\lambda_i > 0$ with $g_i < 0$: a transient after a constraint stops being active; the clamp drives it to zero
- Theory: a fixed μ above a problem-dependent threshold suffices
- Practice: growth speeds convergence (RAI's documented default: slide 24)

Toussaint 2014, arXiv:1412.4329

Backup · The solver landscape, and the saddle-point view

method	idea
augmented Lagrangian	guess the multipliers, minimize, correct them from the residual
primal-dual Newton	Newton on the KKT system in $(\mathbf{z}, \boldsymbol{\lambda}, \boldsymbol{\kappa})$ jointly; interior-point solvers (IPOPT)
sequential quadratic programming	a quadratic program (QP) with the linearized constraints per step; multipliers inherited from the QP (SNOPT)
active-set	guess the touched fences, solve the equality problem, revise

